Clinical Study Summary Report (CSSR) Document Status: Draft | **Version:** 1.0 | **Date:** 03-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A Randomised, Double-blind, Parallel Group, Placebo Controlled, 4-Week, Phase II Study to Evaluate the Effect of AZD4604 on Airway Inflammation and Biomarkers in Adults with Asthma **Short Title/Acronym:** ARTEMISIA **Protocol Number:** D8210C00005 **NCT Number:** NCT06435273 **Sponsor Name:** AstraZeneca **Investigational Product:** AZD4604 (Londamocitinib) / Placebo **Phase of Development:** Phase II **Medical Monitor:** Emmanuel Amadi (AstraZeneca Clinical Study Information Center)

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the effect of AZD4604 compared with placebo on airway inflammation and JAK1-associated signalling pathways in adults with moderate-to-severe asthma over a 4-week treatment period. **Secondary Objectives:** To assess the effects of inhaled AZD4604 on Type 2 (T2) and non-T2 driven inflammatory pathways, evaluate candidate biomarkers of treatment response, and assess safety and tolerability.

2.2 Background & Rationale Patients with uncontrolled asthma despite the use of inhaled corticosteroids (ICS) may have a variety of biological pathways driving their airway inflammation. AZD4604 (londamocitinib), a selective, inhaled, Janus kinase 1 (JAK1) inhibitor, has been designed to target a broad inflammatory cytokine profile including those classically unresponsive to ICS. Because AZD4604 is inhaled, it specifically targets the airways, potentially offering robust efficacy while minimizing systemic side effects typically associated with oral JAK inhibitors.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled, Parallel Group **Blinding:** Double-blind **Randomization:** Randomised (2:1 ratio to AZD4604 or placebo, stratified by baseline FeNO levels above/below 25 ppb)

3.2 Planned Enrollment & Duration Target N: ~36 evaluable adult patients **Number of Sites:** Multicentre (Europe/Global) **Estimated Study Start:** 05-Aug-2024 **Estimated Primary Completion:** 15-Oct-2025

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years, body weight ≥ 40 kg and BMI < 35 kg/m2. 2. Documented history of ≥ 1 severe asthma exacerbation within 1 year prior to Visit 1 (requiring hospitalization/treatment for ≥ 24 hours) OR an Asthma Control Questionnaire-6 (ACQ-6) score ≥ 1.5 at Visit 1. 3. Receiving background treatment with medium-to-high dose inhaled corticosteroid-long-acting beta-agonist (ICS-LABA). 4. Morning pre-BD FEV1 $\ge 60\%$ predicted at Visit 1 and willing/able to undertake bronchoscopy.

4.2 Exclusion Criteria 1. A severe asthma exacerbation within 8 weeks prior to Visit 1. 2. Clinically important pulmonary disease other than asthma (e.g., active lung infection, COPD, bronchiectasis, pulmonary fibrosis, lung cancer, cystic fibrosis). 3. History of herpes zoster reactivation (shingles). 4. Recent use of prohibited medications (e.g., anticoagulants, live/mRNA vaccines, or biologics).

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Administered twice-daily over a 4-week treatment period. **Route of Administration:** Inhalation **Duration of Treatment:** 4 weeks of Investigational Medicinal Product (IMP) administration (Total study duration is approximately 10 weeks).

5.2 Concomitant & Prohibited Therapy Permitted: Background medium-to-high dose ICS-LABA; stable maintenance dose allergen-specific immunotherapy. **Prohibited:** Systemic corticosteroids, biologics/immunoglobulin therapy within 4 weeks, oral JAK inhibitors, live or mRNA vaccines within 4 weeks, anticoagulants, and the use of ICS plus fast-acting β2 agonist as a maintenance and reliever therapy (e.g., Symbicort MART) 30 days prior to Visit 1.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to Day 0	Informed Consent, Medical History, Spirometry, FeNO measurement, First Bronchoscopy
Baseline	Day 0	Randomization (2:1), First Dose of AZD4604 or Placebo
Interim	Weeks 1-3	IMP administration, ePRO diary completion, Safety/Efficacy Labs
End of Treatment	Week 4 (Visit 6a)	Second Bronchoscopy, final biomarker sampling, IP Accountability
Follow-Up	Weeks 5-10	Off-treatment safety follow-up, Exit Interview

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Effect on airway inflammation and JAK1-associated signalling pathways (e.g., target engagement indicators such as phosphorylated STATs) from baseline to the end of the 4-week treatment period. **Measurement Method:** Extensive parallel bio-sampling of the lung target tissue via bronchoscopy.

7.2 Secondary & Exploratory Endpoints * Reduction in Fractional exhaled Nitric Oxide (FeNO). * Changes in lung function (FEV1), asthma symptoms, and ACQ-6 scores. * Multi-omic analysis (transcriptomic, proteomic, metabolomic) of candidate biomarkers in the nasal mucosa, blood, and urine. * Safety and tolerability assessments (Incidence of AEs/SAEs).

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via onsite physical exams, continuous ePRO diaries, vital signs, and central laboratory testing (including liver enzymes and platelet counts). **Serious Adverse Events (SAEs):** Standard reporting timelines for severe asthma exacerbations, signs of systemic JAK inhibition (e.g., infections, shingles), or significant ECG/cardiovascular abnormalities. **Data Monitoring Committee (DMC):** Internal safety review managed by AstraZeneca to monitor the safety of the investigational product and the invasive bronchoscopy procedures.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary biomarker and efficacy endpoints; Safety Set for all adverse events. **Sample Size Justification:** Approximately 36 subjects, powered to detect a mechanistic difference in airway inflammatory signatures between AZD4604 and placebo. **Handling of Missing Data:** Participants discontinuing the study prior to the completion of Visit 6a (the second bronchoscopy) will be actively replaced to ensure sufficient paired biological samples for the primary analysis.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite visits accompanied by centralized data review of biomarker and lung sampling parameters. **Audit Trail:** Secure electronic data capture (eCRF), tracking of electronic patient-reported outcomes (ePRO) device compliance, and formalized data clarification forms.

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11. Study Identifiers & Governance

Primary ID: D8210C00005 **Registry Number:** NCT06435273 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** ARTEMISIA **Official Regulatory Title:** A Randomised, Double-blind, Parallel Group, Placebo Controlled, 4-Week, Phase II Study to Evaluate the Effect of AZD4604 on Airway Inflammation and Biomarkers in Adults with Asthma

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12. Clinical Framework (Asthma Specific)

Primary Condition: Moderate-to-severe Asthma **Diagnosis Threshold:** ≥ 1 severe asthma exacerbation in the prior year or ACQ-6 ≥ 1.5; Morning pre-BD FEV1 $\ge 60\%$ predicted. **Medical Methodology:** Targeted inhalation therapy (selective JAK1 inhibition) to address both Type 2 and non-Type 2 driven airway inflammation. **Optimization Target:** Reduction in JAK1-associated signalling pathways and associated downstream inflammatory markers.

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13. Study Procedures & Methodology

Management Pathway: Standard maintenance ICS-LABA + Investigational inhaled JAK1 inhibitor (or placebo). **Education Protocol (HCP):** Specialized training on parallel bio-sampling protocols and absolute contraindications for the bronchoscopy procedures. **Education Protocol (Patient):** Training on inhaler technique, proper use of the ePRO device for daily symptom tracking, and home spirometer instructions. **Quality Audit Mechanism:** Active replacement of patients missing the second bronchoscopy; routine monitoring of FeNO stratification balance at randomization.

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14. Observation & Follow-Up Schedule

Total Duration: ~10 weeks (4-week IMP exposure + safety follow-up) **Onsite Visit Cadence:** Baseline, Week 4 (Visit 6a/ Bronchoscopy), and designated screening/follow-up visits. **Remote Monitoring Frequency:** Continuous daily ePRO (symptoms and daily metrics). **Checklist Review Interval:** Routine review at every scheduled onsite/interim visit during the 4-week active treatment period.

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15. Sign-off & Approval

| Role | Name | Signature | Date | | :--- | :--- | :--- | :--- | | **Principal Investigator** | [Name] | ____ **| [DD-MMM-YYYY] | |**

Clinical Operations Lead | Emmanuel Amadi | ___ | ***[DD-MMM-YYYY]*** **|**
| ***Lead Statistician*** **|** ***[Name]*** **|** _____ | [DD-MMM-YYYY] |